

Guidelines for the Hackathon Presentation Round

The Hackathon Presentation round will be held on **06th July 2026** at the **Police Control Room, SP Office, City Centre, Gwalior, starting at 12:00 PM onwards.**

Please review the following important details:

1. **Presentation Time:**
 - Each team will be allotted a maximum of **15 minutes** to present their ideas.
2. **Registration:**
 - Only the **first 50 registrations** will be accepted for participation.
 - Participants must enrol on or before 03rd July 2026.
3. **Theme:**
 - All presented ideas must be **strictly related to Cyber Security practices.**

Top ideas might be considered for further prototyping and scaling at the state level.

Problem Statements

01- Problem statements: Develop a software solution to identify active Telegram, WhatsApp, and Instagram channels or bots involved in drug sales in India, and trace users through IP addresses, mobile numbers, and email IDs.

Description: WhatsApp and Telegram channels and Instagram handles are created by drug traffickers to offer various drugs for sale to their subscribers. Customised Telegram bots are also created by some of the drug traffickers to sell drugs. It is most worrisome that the majority of the drugs which are being offered for sale through Telegram, WhatsApp and Instagram are dangerous synthetic drugs like MDMA, LSD, Mephedrone, etc. The above three apps are also used by drug traffickers for drug communication. Expected Solution: Development of a software solution to identify live Telegram and WhatsApp channels/bots and Instagram handles that are offering drugs for sale in India. Solution also should focus on triangulating identifiable parameters like IP address, mobile number, email id etc of the users behind the channel/bot/handle.

Key Tasks :

- **Study and Understand Platform Usage:**
 - Analyze how traffickers use Telegram, WhatsApp, and Instagram to sell drugs (channels, bots, group chats, stories, posts).
- **Create a Keyword and Content Database:**
 - Build a dynamic list of keywords, hashtags, slang, emojis, and phrases related to drug sales (example: MDMA, LSD, party pills, etc.).
- **Real-Time Monitoring Tool:**
 - Develop a tool that continuously scans and monitors Telegram channels, WhatsApp public groups (if accessible), and Instagram handles based on suspicious content.

- **Content Analysis Engine:**
Use AI/NLP to automatically analyze posts, messages, and bot responses to detect drug-related content.
- **Identify Bots and Automation Patterns:**
Detect customized Telegram bots used for automatic drug selling.
- **Extract Metadata for Triangulation:**
Build a system to extract or infer IP addresses, mobile numbers, email IDs, and device fingerprints where possible.
- **Link Accounts to Real-World Identities:**
Correlate different identifiers to try to trace and link online accounts back to real-world users.
- **Dashboard for Investigators:**
Create an easy-to-use dashboard showing flagged accounts, suspected drug-selling activity, and related metadata.
- **Alert Mechanism:**
Integrate an alert system for law enforcement when a new suspicious channel, bot, or handle is detected.
- **Ensure Legal Compliance:**
Ensure that the system respects privacy laws and operates within the boundaries of Indian legal frameworks.

02- Problem statements: Develop a system to identify and track individuals responsible for hoax bomb threats using digital communication channels, such as social media and email, enabling swift action and enforcement to prevent future incidents.

Description: The proposed system aims to identify and track individuals responsible for hoax bomb threats communicated via digital platforms such as social media, email, and messaging services. By leveraging advanced data mining, natural language processing, and digital forensics, the system can detect threatening language, trace sender metadata, and analyze communication patterns. It integrates IP tracking, device fingerprinting, and behavioral profiling to uncover the source of threats, even when attempts are made to anonymize identities. This tool will assist law enforcement in swiftly verifying threats, identifying perpetrators, and taking timely action, ultimately deterring future incidents and enhancing public safety and national security.

Key Tasks:

- **Study Hoax Bomb Threat Patterns:**
Analyze past incidents to understand common language, methods, and digital traces left behind.
- **Build a Threat Language Database:**
Create a list of keywords, phrases, and linguistic patterns commonly associated with bomb threats.
- **Develop Real-Time Monitoring Tools:**
Build tools to monitor emails, social media, and messaging platforms for suspicious or threatening language.

- **Implement Natural Language Processing (NLP):**
Use NLP techniques to automatically detect and flag threatening communications.
- **Metadata Extraction and Analysis:**
Design a system to extract sender IP addresses, device information, timestamps, and other metadata from communications.
- **Apply Device Fingerprinting:**
Use device fingerprinting methods to track devices even when users change accounts or attempt to stay anonymous.
- **Behavioral Profiling:**
Analyze communication habits (time, style, frequency) to build behavior profiles and link multiple threats to the same individual.
- **Link Digital Identities to Real Persons:**
Correlate metadata and profiles to identify the real-world identities behind anonymous threats.
- **Build a Dashboard for Investigators:**
Provide a user-friendly dashboard showing flagged threats, related metadata, and risk levels for easy action by law enforcement.
- **Ensure Legal and Ethical Compliance:**
Ensure the system works within privacy laws, data protection guidelines, and cybercrime regulations.

03- Problem statements: Develop a smart tool to extract and identify B-party (recipient)Public IP/ mobile numbers from IPDR logs, enabling accurate mapping of A-party to B-party interactions for use in law enforcement investigations.”

Description: The proposed problem involves developing an intelligent system to efficiently extract and identify B-party (recipient) mobile numbers/Public IP from Internet Protocol Detail Records (IPDR). This tool should be capable of parsing large and complex log files, filtering relevant communication data, and accurately mapping interactions between A-party (initiator) and B-party users. The goal is to support law enforcement, telecom investigations, and security analysis by providing clear insights into communication patterns. The system must ensure high accuracy, handle diverse IPDR formats, and streamline investigative workflows, enabling faster identification of suspects, networks, or suspicious activity in digital communication data.

Key Tasks :

- **Understand IPDR Structure and Formats:**
Study different types of IPDR logs from various telecom operators to understand how data is recorded.
- **Develop a Log Parsing Engine:**
Build a system that can read, parse, and clean large and complex IPDR files.
- **Identify A-Party and B-Party Relationships:**
Extract and map initiator (A-party) and recipient (B-party) numbers/ IPs accurately from the logs.

- **Filter Relevant Communication Data:**
Design filters to exclude irrelevant records and focus on communication sessions useful for investigation.
- **Normalize Diverse Data Formats:**
Ensure the system can handle different file types and formats (CSV, TXT, JSON, etc.) across various telecom providers.
- **Build Communication Mapping Tools:**
Create visual graphs or tables showing how different numbers are connected through communication patterns.
- **Automate Detection of Suspicious Activity:**
Add pattern analysis to automatically highlight unusual communication behaviors (e.g., frequent late-night calls, short-duration calls).
- **Create a Search and Query System:**
Allow investigators to search for specific numbers, Public IP, date ranges, and communication types easily.
- **Develop a User-Friendly Dashboard:**
Design a simple interface to display extracted numbers, call relationships, and potential red flags.
- **Ensure Data Security and Compliance:**
Implement strong security measures to protect sensitive communication data and comply with privacy laws.

04- Problem statements: Develop a solution to trace VoIP calls by analyzing network traffic, metadata, and signaling protocols, even with encryption, to help law enforcement track suspicious or malicious communications.

Description: The solution aims to trace Voice over IP (VoIP) calls by analyzing network traffic, metadata, and signaling protocols such as SIP and RTP, even when encryption is present. By leveraging packet inspection, flow analysis, and timing patterns, the system can detect call initiation, duration, endpoints, and communication frequency without accessing the call content. It correlates metadata with known IP addresses, user agents, and call behaviors to identify suspicious activity. Designed for law enforcement, the tool enhances surveillance capabilities, enabling the tracking of malicious or unauthorized VoIP communications while preserving data privacy and adhering to legal investigation standards.

Key Tasks :

- **Understand VoIP Protocols:**
Study key VoIP protocols like SIP (Session Initiation Protocol) and RTP (Real-Time Transport Protocol) to understand call setup, control, and data flow.
- **Develop Packet Inspection Tools:**
Create tools for deep packet inspection (DPI) to extract metadata even from encrypted VoIP traffic.
- **Perform Flow and Timing Analysis:**
Analyze network flows and timing patterns (such as packet sizes and intervals) to detect when a VoIP call starts and ends.

- **Capture and Analyze Metadata:**
Extract metadata like caller IP addresses, device information, call duration, and call frequency without decrypting voice content.
- **Correlate with Known Databases:**
Match captured IP addresses, user agents, and communication patterns with known blacklists or suspected malicious entities.
- **Identify Anomalous Communication Behavior:**
Detect unusual calling patterns (frequent short calls, communication with high-risk regions, hidden servers) to flag suspicious activity.
- **Maintain Privacy Compliance:**
Ensure the system focuses only on metadata and adheres strictly to legal surveillance and privacy standards.
- **Build a Real-Time Monitoring Dashboard:**
Develop an intuitive dashboard to visualize active VoIP calls, suspicious users, and risk scores for law enforcement use.
- **Implement Alert Mechanisms:**
Set up automated alerts for detected suspicious VoIP activity, unusual behavior, or blacklisted IP communications.
- **Test in Simulated and Real Environments:**
Validate the system's effectiveness using both simulated VoIP networks and real-world anonymized datasets.

05- Problem statements: Develop a system to detect anti-India campaigns on digital platforms by building a keyword database and using engagement analysis to identify key drivers, supporting early detection.

Description: The proposed system aims to detect anti-India campaigns on digital platforms by combining keyword-based monitoring with engagement analysis. It involves creating a dynamic database of flagged terms, phrases, and hashtags associated with anti-national sentiments. Using AI and natural language processing, the system scans social media, forums, and websites to identify content matching the database. It then analyzes user engagement—likes, shares, comments, and network reach—to pinpoint influential accounts and viral trends. This approach enables early detection of coordinated disinformation or propaganda efforts, supporting timely countermeasures by security agencies and helping preserve national integrity in the digital space.

Key Tasks:

- **Build a Keyword and Phrase Database:**
Collect and maintain a dynamic list of terms, hashtags, and phrases linked to anti-India sentiments or propaganda.
- **Develop a Real-Time Content Monitoring Tool:**
Create tools to scan social media platforms (like Twitter, Facebook, Instagram), forums, and websites for matching content.
- **Implement Natural Language Processing (NLP):**
Use NLP techniques to understand and filter out relevant anti-India content from large volumes of text data.

- **Conduct Engagement Analysis:**
Analyze metrics like likes, shares, retweets, comments, and reach to identify which posts and accounts are gaining traction.
- **Identify Key Influencers and Networks:**
Pinpoint influential users, pages, or groups spreading anti-India content and map their network of followers or supporters.
- **Detect Coordinated Campaigns:**
Recognize signs of organized efforts, such as similar posts by multiple accounts at the same time or coordinated hashtag usage.
- **Develop an Alert System:**
Set up real-time alerts for trending anti-India narratives or when engagement crosses a set threshold.
- **Create a Visualization Dashboard:**
Design a dashboard showing flagged content, engagement statistics, influential accounts, and campaign networks.
- **Ensure Legal and Ethical Compliance:**
Make sure that monitoring is conducted within the limits of free speech and privacy laws.
- **Support Countermeasure Strategies:**
Provide actionable insights and reports to government and security agencies for taking preventive or corrective action.

06- Problem statements: Undertake research, on analysis of social media space with particular focus on algorithm employed by social media platform.

A research project on social media algorithms could explore how they shape user experiences, influence information consumption, and impact society, examining both positive and negative consequences.

Here's a breakdown of potential research areas and questions-

A Understanding the Algorithms:

- What are social media algorithms?
- Define algorithms as the automated processes that determine content visibility and user experiences.
- How do different platforms use algorithms?
- Compare and contrast the algorithms of platforms like Facebook, Instagram, TikTok, Twitter, and YouTube.
- What factors influence algorithm decisions?
- Investigate how user interactions, content types, and other signals impact content ranking and recommendations.
- How do algorithms learn and adapt?
- Explore the use of machine learning and how algorithms evolve based on user data and feedback.

B Impact on User Experience:

- How do algorithms affect user engagement?
- Analyze the impact of algorithms on time spent on platforms, likes, shares, and comments.
- Do algorithms create "echo chambers" or filter bubbles?
- Investigate how algorithms can limit exposure to diverse perspectives and viewpoints.
- How do algorithms influence user perceptions and opinions?
- Examine the potential for algorithms to shape beliefs and attitudes.
- What is the impact on mental health and well-being?
- Explore the potential for algorithms to contribute to anxiety, depression, and other negative psychological effects.

C Societal Impact:

- How do algorithms facilitate the spread of misinformation and fake news?
- Investigate the role of algorithms in amplifying false or misleading content.
- What is the impact on social polarization and division?
- Analyze how algorithms can reinforce existing biases and create divisions within society.
- How do algorithms affect access to information and knowledge?
- Explore the potential for algorithms to limit access to diverse perspectives and information sources.
- What are the ethical implications of algorithms in social media?
- Discuss issues of privacy, bias, and manipulation.